

3.Git-HOL

What is Branching?

A **branch** is a separate copy of your project's code where you can work on new features or fix bugs **without affecting the main project**.

By default, every Git repository has a branch called **main** (or **master** in older repositories).

Creating a new branch allows multiple developers to work on different features at the same time.

Why do we use branches?

- To develop new features safely.
- To fix bugs without affecting the main code.
- To allow multiple developers to work simultaneously.
- To test changes before adding them to the main branch.

Example

Suppose you are building a Resume Builder project.

- **main** → Stable version of the project.
- **login-feature** → Adding a login page.
- **resume-template** → Creating a new resume template.
- **bug-fix** → Fixing a validation error.

Each developer can work on their own branch independently.

Creating a Branch

Create a new branch:

```
git branch feature-login
```

View all branches:

```
git branch
```

Switch to the new branch:

```
git checkout feature-login
```

Or create and switch in one command:

```
git checkout -b feature-login
```

What is Merging?

Merging means combining the changes from one branch into another branch.

After completing and testing your work on a feature branch, you merge it into the **main** branch.

Example

Before merging:

main

|

|----- feature-login

After merging:

main (contains login feature)

Steps to Merge a Branch

Step 1: Switch to the main branch

```
git checkout main
```

Step 2: Merge the feature branch

```
git merge feature-login
```

Step 3: Push the updated main branch

```
git push origin main
```

Benefits of Branching and Merging

- Keeps the main branch stable.
- Allows parallel development.
- Makes teamwork easier.
- Helps review and test code before merging.

- Makes it easier to manage different features and bug fixes.
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Creating a Branch Request in GitLab

A **branch request** generally means creating a **new branch** in a GitLab repository so that you can work on changes without modifying the main branch.

Steps

1. Open your GitLab project.
2. Go to **Repository → Branches**.
3. Click **New Branch**.
4. Enter a branch name (for example, feature-login).
5. Select the source branch (usually main).
6. Click **Create Branch**.

Now you can make changes on this new branch and push your commits to it.

Creating a Merge Request (MR) in GitLab

A **Merge Request (MR)** is used to request that the changes from one branch be merged into another branch (usually main).

It allows teammates to review your code before it becomes part of the main project.

Steps to Create a Merge Request

1. Push your branch to GitLab.

`git push origin feature-login`

2. Open your GitLab repository.
3. Go to **Merge Requests**.
4. Click **New Merge Request**.
5. Select:
 - **Source branch:** feature-login
 - **Target branch:** main
6. Click **Compare branches and continue**.
7. Add a title and description explaining your changes.

8. Click **Create Merge Request**.
9. Team members review the code.
10. If everything looks good, click **Merge** to merge the changes into the target branch.

Why Use Merge Requests?

- Code is reviewed before merging.
- Helps find bugs early.
- Encourages collaboration.
- Keeps the main branch stable.
- Maintains a history of discussions and changes.